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1. (a) Prove by induction that, for all integers $n \geq 1$

$$\sum_{r=1}^n r^3 = \frac{1}{4}n^2(n+1)^2 \quad [4]$$

- (b) Hence, or otherwise, write down a factorised expression for the sum of the first $2n$ cubes

$$1^3 + 2^3 + 3^3 + \cdots + (2n)^3 \quad [1]$$

- (c) Use the formula in part (a) to write down a factorised expression for the sum of the first n even cubes

$$2^3 + 4^3 + 6^3 + \cdots + (2n)^3 \quad [2]$$

- (d) Hence, or otherwise, show that

$$1^3 + 3^3 + 5^3 + \cdots + (2n-1)^3 = an^2(bn^2 - 1)$$

where a and b are rational numbers to be determined. [3]

2. Prove by induction that, for $n \geq 1$

$$\sum_{r=1}^n r 2^r = 2 + (n-1)2^{n+1}$$

[5]

3. Prove by induction that for all positive integers n

$$\sum_{r=1}^n \log \left(1 - \frac{1}{4r^2} \right) = \log \left(\frac{(2n)!(2n+1)!}{2^{4n}(n!)^4} \right) \quad [6]$$

4. Prove by induction that, for $n \in \mathbb{Z}^+$,

$$\sum_{r=1}^n \frac{1}{r(r+1)(r+2)} = \frac{n(n+3)}{4(n+1)(n+2)} \quad [5]$$

5. Prove by induction that, for $n \geq 1$

$$\sum_{r=1}^n (2^r + 2 \times 3^r) = 2^{n+1} + 3^{n+1} - 5$$

[5]

6. Prove by induction that for $n \in \mathbb{Z}^+$

$$\sum_{r=1}^n r3^{r-1} = \frac{1 + (2n-1)3^n}{4}$$

[6]

7. (a) Prove by induction that, for all positive integers n

$$\sum_{r=1}^n \frac{1}{(2r-1)(2r+1)} = \frac{n}{2n+1} \quad [6]$$

(b) Hence, show that, for all positive integers n

$$\sum_{r=n+1}^{2n} \frac{1}{(2r-1)(2r+1)} = \frac{n}{(an+b)(cn+d)}$$

where a, b, c and d are integers to be determined.

[3]

8. Prove by induction that, for $n \geq 1$

$$\sum_{r=1}^n \frac{r}{2^r} = 2 - \frac{n+2}{2^n}$$

[5]

9. (a) Prove by induction that for all positive integers n

$$\sum_{r=1}^n (2r-1)^3 = n^2(2n^2-1) \quad [6]$$

(b) Given that

$$\sum_{r=1}^{2n} (2r-1)^3 = k \sum_{r=1}^n (2r-1)^3$$

show that

$$n^2 = \frac{k-4}{2(k-16)} \quad [5]$$

10. Prove by induction that for all positive integers n ,

$$\sum_{r=1}^n r(r+1)(r+2) = \frac{1}{4}n(n+1)(n+2)(n+3)$$

[6]